Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Modularized SMR Simulator for Cloud Computing Environment			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power (√), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Energy & Resources			
	Medium Category	Small Category	Medium Category	Small Category
	Nuclear Power	Next-Generation Nuclear Power		
Concept of Proposed Technology	Various components including nuclear reactor, thermal hydraulic heat transfer mechanism, thermodynamic process, electromagnetic generators, mechanical actuators, and control logics of Small modular reactors (SMR), including light-water reactors and advanced GenIV reactors will be modeled by using Modelica simulation language. Developed component models will be connected to each other in a cloud computing environment to simulate the behavior of various options of nuclear plants.			
Rationale for Joint R&D Project	Effective mathematical simulation relies on user-friendly interfaces and efficient resource management—both of which are provided by cloud computing. Leveraging RPI's specialized Modelica simulation models for nuclear reactors, this cloud-based approach enables much advanced analysis for more robust design V&V			
Key Words	1. SMR design V&V 2. Cloud computing 3. Modularized simulation			

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (6)

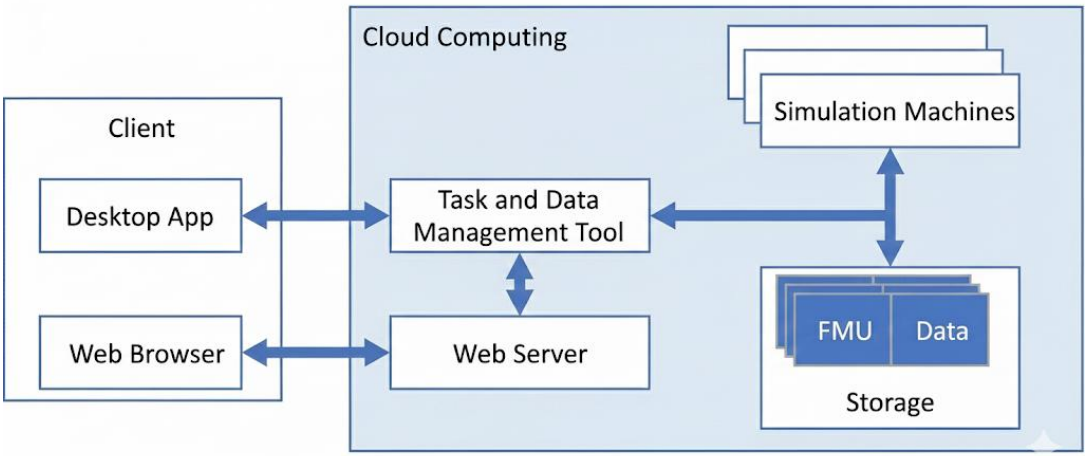
R&D Objectives

- Develop modularized Modelica simulation components (typical LWR reactors, GenIV reactors, pumps, valves, heat exchangers, turbines, generators, control logic, etc.)
- Verify the developed components by comparing them with existing credible models like RELAP or MARS.
- Develop a cloud-based platform and its user interface to perform integrated analysis and establish an SMR task registration and management system for systematic process oversight
- Develop intuitive, graphics-based task component nodes and high-performance solvers
- Develop a SMR design V&V platform with a scalable digital engineering capability and cloud environment.

R&D Target Specifications

- Constructing more than 20 variations of SMR design by combining the developed components
- Simulating SMR plant behavior in transient operation at real-time or faster speed
- Integrated analysis with more than 100 simulation agents in cloud environment

R&D Plan



<Conceptual configuration of developed platform>

Approach

- This R&D project aims at the development of modularized SMR simulator for cloud computing environment. Modelica language will be used for simulation component development since it provides capability of handling different time step simulations. Dymola (Dassault Systems) programing environment will be utilized for the enhanced user interface and the extended library. TRANSFORM library (Oak Ridge National Lab) will be used for the nuclear reactor core models while OpenIPSL library (RPI) will be used for electrical components. In-house developed existing models for various nuclear reactors and energy transfer components will also be used.
- The final component models will be systematically verified against the results from well-

accepted codes like RELAP or MARS. Once verified, the Modelica models will be converted into C source codes to be complied into various forms of modularized component models.

- For the integrated simulation, dedicated user interface and task management platform will be developed. The simulation of a plant which consists of multiple components will be optimized for cloud computing environment.
- User interface can be web-based or dedicated application-based. Through this user interface, multiple variations of possible SMR options should be created and tested. This interface agent is separated from nuclear plant simulation elements.
- For the systematic V&V of the SMR design options, multiple simulation agents will be mobilized in cloud computing environment to scale simulation performance beyond the capabilities of a single computer, and their results will be collected and analyzed in the developed platform.

Tasks

Task 1: Development of SMR component models

- Modelica language will be used for simulation.
- In-house component models, open sources for general components, and public or commercial library will be fully used.
- Various size and capacity of multiple components like reactor cores, heat exchangers, valves, pumps, tanks, steam generators, turbines, and electricity generators will be addressed
- Each component model will get its modeling description for user reference

Task 2: Verification with transient responses

- Typical test transient events must be pre-determined for various reactor types considering characteristics of plant responses
- Verification reference models for selected components will be developed by using well-accepted industry codes like RELAP and MARS.
- Simulation results of Modelica model of each component option for transient events will be compared to the verification references
- The Modelica models will be adjusted to make them well-aligned to the references

Task 3: User Interface for constructing an integrated model

- Innovated modeling and analysis productivity by developing graphic-based intuitive task component nodes and high-performance solvers.
- Enhanced collaboration efficiency across the entire design and operation lifecycle by building graphic-based drawing and node-driven interaction services.
- Advanced real-time monitoring and operation support by implementing operator-friendly HMI editors and runtimes.

Task 4: Task management platform

- Constructed models will be simulated under specified initiating events and transients
- Operational transients will be created, stored, and imposed to the specific target simulation model, and their results will be stored in a coded format.
- Implementations must be aligned with the architectural styles of cloud computing applications

	Task 5: Porting to Cloud computing environment - Developed models, user interface, and task management tool will be ported to Cloud computing environment - By using industry standard Functional Mock-up Interface, Functional Mockup Units will be produced for co-simulation as an executable form of a model - Self-executing simulation units will be deployed on multiple machines/agents to achieve combined cloud computing and system simulation. - This task will deliver a highly scalable digital engineering platform for SMR design verification by establishing a cloud-based simulation and analysis environment. Task 6: Result analysis tool for design V&V - By using the task management platform in the cloud computing environment, large number of simulations can be executed in a short time, and their result will be collected. - A tool that performs the analysis of results in consideration of input variation in a systematic manner will be developed - Insightful parameters and characteristic features should be extracted automatically based on the instruction of system designer. Task 7: Pilot applications - In the pilot applications, Sodium cooled reactor variations and light-water-based SMR variations will be addressed. - Engineering insights for enhanced plant design in terms of both safety and operability will be produced from the series of pilot applications. <u>Deliverables with Timeline</u> Deliverable 1: Set of Modelica simulation models for the components of SMRs (Year 1) Deliverable 2: User interface software for integrated simulation of SMRs (Year 1) Deliverable 3: Verification report of simulation components (Year 1.5) Deliverable 4: Task management platform software (Year 2) Deliverable 5: Finalized cloud computing environment (Year 2.5) Deliverable 6: Pilot application report (Year 3) Deliverable 7: Final report (Year 3)

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	RPI (PI: Hyun Gook Kang) will be leading efforts for Deliverables 1, 3, and 6. - More specifically, RPI will develop Modelica simulation models for every key component of various SMR options. - Simulation libraries, in-house models, and other available resources will be mobilized. - Main target SMRs are light-water-based reactors and sodium-cooled reactors. - Development of verification reference models by using RELAP or MARS will be performed. - Verification report of Modelica simulation models will be produced. - The pilot application of the developed Cloud-computing Modelica simulation for a couple of target SMRs will be performed. - Target SMR design verification based on the transient response analysis results in Cloud-computing environment be performed. Deliverable 7 will be collaboratively produced by integrating all activities of this R&D effort.
Expected R&R of Korean SME Partner	Korean companies are expected to lead efforts for Deliverables 2, 4, and 5. - Graphical user interface for constructing a SMR design based on provided component library will be developed and tested. - Task management platform that manages operational transients for target simulation models will be developed. - Simulation results must be stored in a coded format so that multiple results can be complied into meaningful design improvement insights. - Porting all the developed models, user interface software and task managing software to Cloud computing environment will be performed. - Data produced by multiple simulation machine/agents should be managed in a unified format for a systematic and integrated analysis. Deliverable 7 will be collaboratively produced by integrating all activities of this R&D effort.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Leveraging the scalability of cloud-based high-performance computing, this project aims to develop a modular, high-fidelity verification platform for Small Modular Reactors (SMRs). By using the Modelica standard, the platform offers an object-oriented modeling environment that allows users to seamlessly integrate and reconfigure diverse reactor components. This flexibility enables the rapid prototyping and safety analysis of various SMR design options, providing critical insights into transient behavior and system-level performance. Ultimately, this platform serves as a basis of standardized, commercial-grade foundation for accelerating the licensing and deployment of next-generation nuclear technologies.

6. References

(1) Kyle Warns, Hyun Gook Kang, “Simulation of Electrical System Event Propagation in Nuclear Power Plants Using Modelica,” NPIC&HMIT 2025, ANS, Chicago, IL, June 2025.

(2) Xingang Zhao, Junyung Kim, Kyle Warns, Xinyan Wang, Pradeep Ramuhalli, Sacit Cetiner, Hyun Gook Kang, Michael Golay “Prognostics and health management in nuclear power plants: an updated method-centric review with special focus on data-driven methods”, Frontiers in Energy Research, Vol.9, 2021

(3) Asad Ullah Amin Shah, Kyle Warns, Junyung Kim, Miguel Aguilera, Luigi Vanfretti, Hyun Gook Kang. “Nuclear power system-level modeling and simulation-Comparing thermal library suites.” 11th Workshop on Modeling and Simulation of Cyber-Physical Energy Systems, San Antonio, TX, May 2023

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization